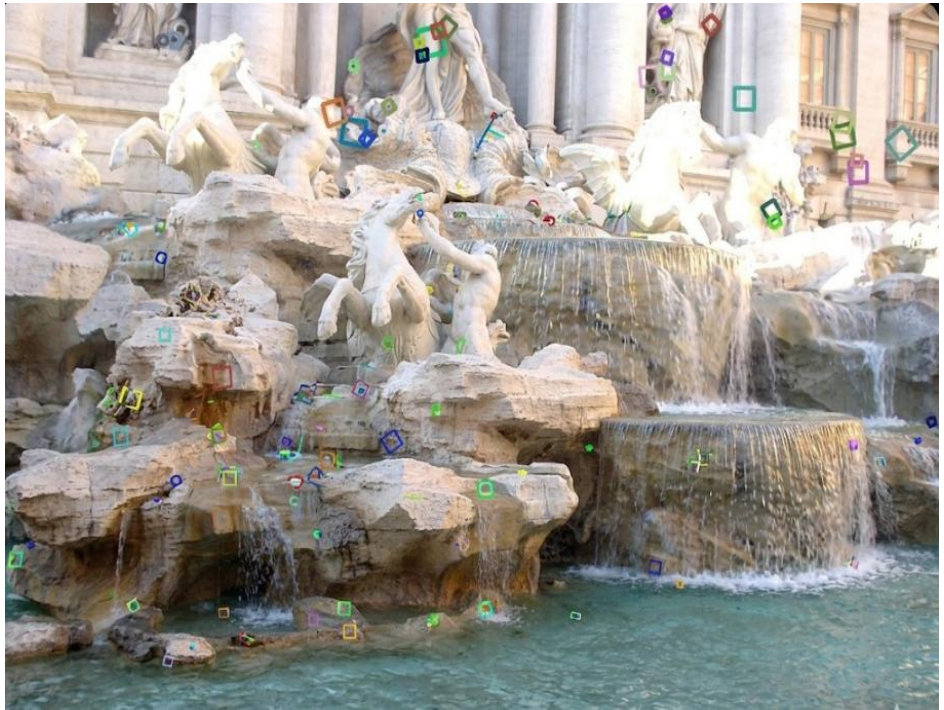
Properties of SIFT

Extraordinarily robust matching technique

- Can handle changes in viewpoint (up to about 60 degree out of plane rotation)
- Can handle significant changes in illumination (sometimes even day vs. night (below))
- Pretty fast—hard to make real-time, but can run in <1s for moderate image sizes
- Lots of code available

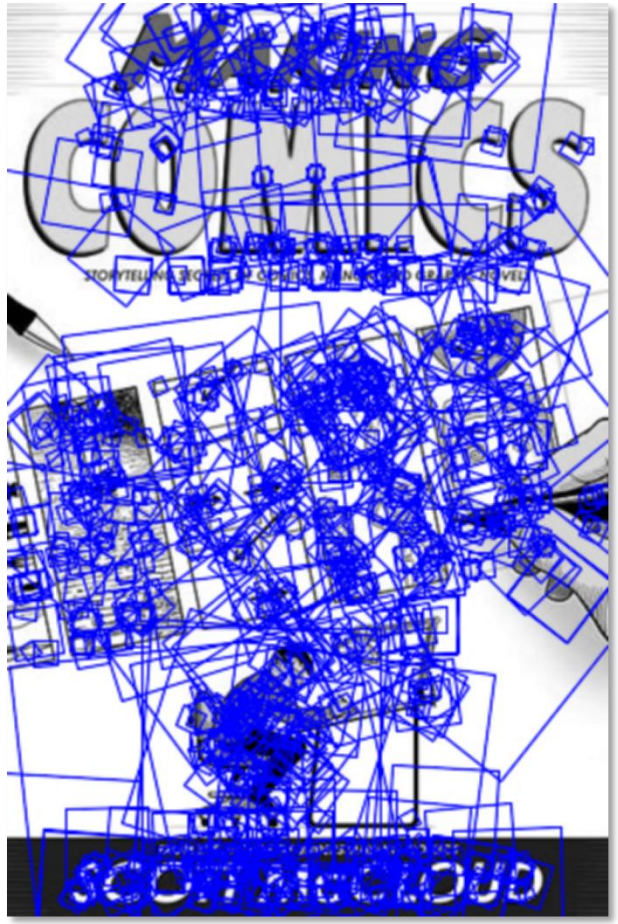




SIFT Example



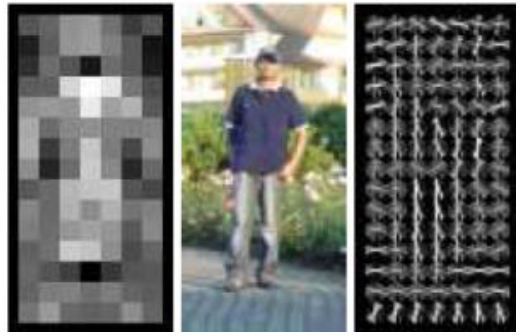
sift



868 SIFT features

Other descriptors

- HOG: Histogram of Gradients (HOG)
  - Dalal/Triggs
  - Sliding window, pedestrian detection



- FREAK (Fast Retina Keypoint) or ORB features
  - Can run in real-time; used in Visual SLAM on-device

- LIFT: Learned Invariant Feature Transform
  - Learned via deep learning – along with many other recent features

https://arxiv.org/abs/1603.09114

Other descriptors

- HOG: Histogram of Gradients (HOG)
  - Dalal/Triggs
  - Sliding window, pedestrian detection

1. Gradient Computation

- Compute horizontal and vertical gradients using [-1, 0, 1] filters

- Calculate gradient magnitude and orientation at each pixel

2. Cell Histograms

- Divide image into small cells (e.g., 8×8 pixels)

- Create histogram of gradient orientations (typically 9 bins, 0-180°)

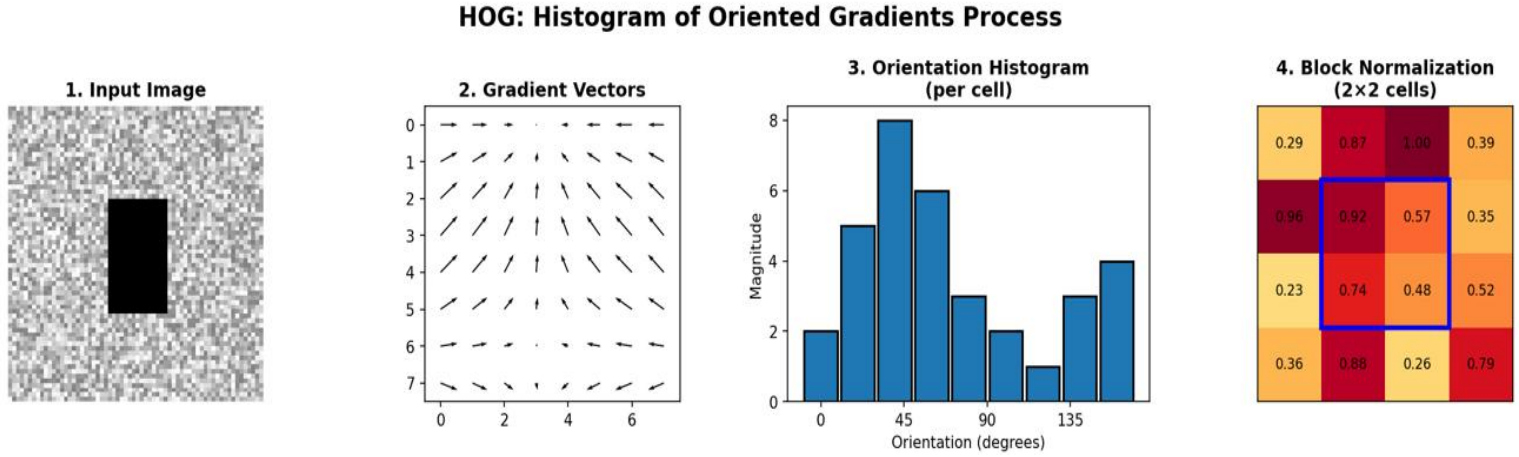
- Weight votes by gradient magnitude

3. Block Normalization

- Group cells into overlapping blocks (e.g., 2×2 cells)

- Normalize histograms within each block (L2-norm)

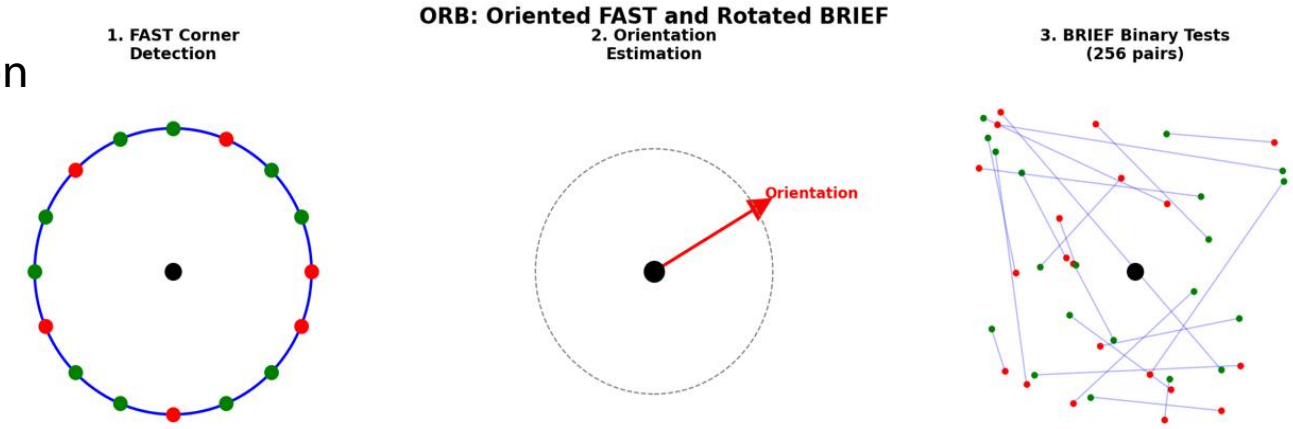
- Provides invariance to illumination changes



Other descriptors

- FREAK (Fast Retina Keypoint) or ORB features
  - Can run in real-time; used in Visual SLAM on-device

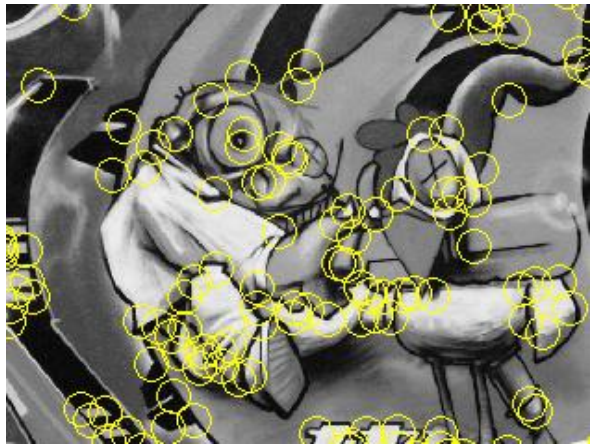
1. oFAST: Oriented FAST Keypoint Detection
   - Uses FAST corner detector to find keypoints
   - Computes intensity centroid to determine orientation
   - Applies Harris corner measure to rank keypoints
2. rBRIEF: Rotated BRIEF Descriptor
   - Binary descriptor: compares intensity of pixel pairs
   - 256 bits (256 binary tests)
   - Rotates sampling pattern according to keypoint orientation
   - Uses learned sampling patterns for better performance



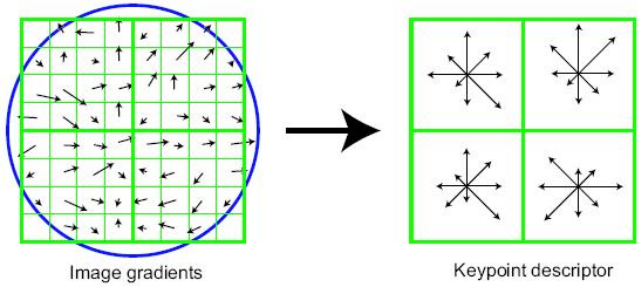
Questions?

Summary

- Keypoint detection: repeatable and distinctive
  - Corners, blobs
  - Harris, DoG



- Descriptors: robust and selective
  - spatial histograms of orientation
  - SIFT and variants are typically good for stitching and recognition
  - But, need not stick to one



Which features match?

